The Software-Defined Vehicle: A Comprehensive Study on Current Trends and Challenges

—JOHANNES STÜMPFLE
Institute of Industrial Automation and Software Engineering, University of Stuttgart, 70569 Stuttgart, Germany

—JOHANNES SIGEL
Institute of Industrial Automation and Software Engineering, University of Stuttgart, 70569 Stuttgart, Germany

—MATTHIAS WEIß
Institute of Industrial Automation and Software Engineering, University of Stuttgart, 70569 Stuttgart, Germany

Graduate Student Member, IEEE

—BARAN CAN GÜL
Institute of Industrial Automation and Software Engineering, University of Stuttgart, 70569 Stuttgart, Germany

—FALK DETTINGER
Institute of Industrial Automation and Software Engineering, University of Stuttgart, 70569 Stuttgart, Germany

—NASSER JAZDI
Institute of Industrial Automation and Software Engineering, University of Stuttgart, 70569 Stuttgart, Germany

Senior Member, IEEE

—MAX HOßFELD
InnovationCampus Future Mobility, University of Stuttgart, 70569 Stuttgart, Germany

—MICHAEL WEYRICH
Institute of Industrial Automation and Software Engineering, University of Stuttgart, 70569 Stuttgart, Germany

(Corresponding author: Johannes Stümpfle.)

The review of this article was arranged by Department Editor Breno Nunes.

IEEE DOI 10.1109/EMR.2025.3636574

Abstract*— *The automotive industry is undergoing an extensive transition from hardware-centric development to software-defined vehicles (SDVs). This paradigm shift requires innovative technical solutions and fundamentally different approaches for vehicle design. Together, these developments introduce substantial challenges for an entire industry, affecting both development processes and operations. Although SDVs have attracted considerable attention and widespread industry involvement, prior research has primarily emphasized systematic literature reviews identifying emerging academic trends. Consequently, little attention has been paid to industry and economic developments that influence how SDV technologies evolve in practice and are crucial for future advancement. In response, this article provides a different, industry-oriented view of the current trends and associated challenges of SDVs. To achieve this, we conducted a comprehensive analysis of the SDV from an industry-based perspective. We analyzed three sources of information to identify the current trends in SDV development: expert interviews with automotive IT specialists, a study on major automotive trade fairs and workshops, and insights gained from the authors' professional experience. Based upon, we derive specific challenges arising from these trends. Our findings offer a comprehensive overview of the present SDV landscape, trends, and challenges, and serve as a foundation for future research efforts in this rapidly evolving field.

Key words: Autonomous vehicle, software-defined vehicle (SDV), survey, trends and challenges.

I. INTRODUCTION

IN RECENT years, the automotive industry has witnessed a significant shift toward software-defined vehicles (SDV), reflecting the increasing integration of software into vehicles [1]. Software innovations are dominating the progress of the modern vehicle, underscoring the transformative nature of this trend [2]. SDV refers to vehicles where much of the functionality traditionally associated with hardware is implemented in software, offering a plethora of benefits including flexibility, adaptability, and enhanced user experiences. The idea of a "smartphone on wheels" asks for the possibility of continuously updating the software of the SDV through over-the-air (OTA) updates to provide the vehicle with new functionalities or bug fixes. However, transitioning from a hardware-centric approach to a software-defined paradigm necessitates a comprehensive and complex transformation. This transformation encompasses the adoption of new software and electrical/electronic (E/E) architectures, novel development methodologies, and innovative edge and cloud strategies that underpin SDV capabilities. Within

0360-8581 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies.
Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Universidad de Malaga. Downloaded on September 13,2026 at 16:03:57 UTC from IEEE Xplore. Restrictions apply.

this infrastructure, the individual vehicle becomes just a small component within a distributed and interconnected network comprising vehicles, edge, and cloud instances [3]. Moreover, the trajectory of SDV development is heavily influenced by industry and economic factors [4]. Historically, the automotive sector has been hardware-centric, emphasizing physical components over software innovation. However, as the landscape shifts toward software-driven solutions [2], the industry must navigate uncharted territory, requiring collaboration between automotive manufacturers, technology companies, policymakers, and other stakeholders. Despite the pivotal role of industry and economic forces, there exists a notable gap in research analysis focusing on current trends and their implications for SDV development. Understanding and leveraging these trends are essential for informing future research agendas and shaping the trajectory of SDV innovation. Therefore, this article aims to provide an overview of the current trends in engineering the SDV, shedding light on key developments and their potential implications for the future of SDV research and development.

Therefore, we conducted a comprehensive analysis on engineering the SDV, utilizing three independent sources: interviews with automotive IT experts, studies of major automotive trade fairs and workshops, and expert knowledge gained from various research projects with high industry participation and attendance at industry workshops and conferences. This analysis led to the identification of 11 current trends in the SDV field. Subsequently, we pinpointed challenges associated with these trends that require attention for further advancement. Finally, we explored future research on SDV, drawing from our industry perspective and the identified trends.

The remaining article is structured as follows. First, a literature review was conducted to give an overview of the basis regarding SDV and its origin and to outline the dimensions of engineering the SDV. In Section III, we describe the methodology used for our analysis to foster the trends. Subsequently, we present our results of our analysis in IV. Based on the results, we outline a discussion in Section V. Finally, a conclusion and an outlook regarding future work are given in Section VI.

II. Literature Review

This literature review examines the current state of academic research within the three key dimensions that emerged from our empirical investigation. Section II-A establishes the foundational concept of the SDV, while the subsequent sections explore the architectural transformations (Section II-B), data management and processing challenges (see Section II-C), and broader economic and political implications (see Section II-D) that characterize the technological shift.

A. Software-Defined Vehicle The automotive industry is undergoing a fundamental transformation driven by the emergence of SDVs, which represent a paradigm shift from traditional mechanical and electrical systems to software-centric architectures. The concept of the "software-defined vehicle" refers to modern vehicles being increasingly controlled and enhanced by software rather than purely mechanical or electrical systems [5]. In this paradigm, key functionalities—from driving controls to infotainment systems and safety features—are primarily governed by software that can be updated and modified over time, similar to computers or smartphones.

This transformation enables continuous updates through DevOps principles, utilizing an IT infrastructure of cloud and edge systems for vehicle communication and data collection. The resulting data loop between development and operation teams facilitates system monitoring for operational faults and software improvements. The motivation for SDVs is multifaceted: automakers can improve vehicle functionalities postsale through over-the-air updates, enhancing customer satisfaction while quickly resolving software bugs and security vulnerabilities. In addition, software enables advanced features such as autonomous driving, personalized driver settings, and connectivity features that integrate vehicles with digital life.

The SDV concept draws from software-defined networking (SDN) [6] and smartphone technology transformation. By separating software from hardware and enabling remote management, similar to SDN's control logic separation, the automotive sector achieves greater flexibility. Virtualization techniques allow dynamic determination of function locations during operation, promising manufacturing and operating cost savings through more resource-efficient operations. This evolution toward isolated software applications with communication features is captured by the "smartphone on wheels" analogy, emphasizing vehicles as connected devices delivering customized, up-to-date services throughout their lifecycle.

B. Architecture The realization of SDVs necessitates a fundamental redesign of ecosystem architecture and infrastructure, transforming the traditional automotive paradigm into an interconnected digital system that extends beyond individual vehicles to encompass the entire mobility landscape [7], [8].

At the heart of this ecosystem transformation lies the digital vehicle architecture, which must seamlessly integrate on-board software and E/E architecture with off-board components, particularly backend architecture. The European

Authorized licensed use limited to: Universidad de Malaga. Downloaded on September 13,2026 at 16:03:57 UTC from IEEE Xplore. Restrictions apply.

Commission's concept paper introduced an initial vision defining on-board and off-board segments with tight interconnection and shared structure between the digital vehicle and its backend [9]. This architecture introduces abstraction layers to encapsulate processor topologies, incorporating hypervisors while providing unified abstractions for higher level vehicle functions, ultimately seeking to define standardized open APIs for seamless integration and interoperability.

The on-board E/E architecture is transitioning from distributed or domain-centralized models with numerous electronic control units (ECUs) to zonal architectures featuring high-performance computers (HPC) grouped into vehicle zones: advanced driver-assistance systems (ADAS), cockpit, powertrain, body and comfort, and connectivity [10]. This architectural evolution facilitates the separation between software and hardware, supporting SDV objectives.

Managing architectural complexity is challenging due to high configurability requirements, market-specific adaptations, and hardware restrictions that create significant variability. This variability exists in space (vast product range) and time (continuous evolution through short development cycles enabled by agile methods, CI/CD pipelines, and OTA updates) [11]. The extensive variability across both spatial and temporal dimensions in SDV development creates significant challenges in managing software complexity and preventing technical debt accumulation [12]. Consequently, there is a pressing need for systematic variability management approaches. Software product line engineering (SPLE) has emerged as a promising solution in the automotive domain, leveraging reusable software artifacts through structured domain engineering and application engineering processes [13], [14]. Despite its potential, SPLE adoption in automotive contexts continues to face implementation and evolution challenges [15]. Recent research has begun exploring the potential of generative artificial intelligence technologies to address these variability management challenges [16], [17].

Increasing vehicle function complexity demands high-performance hardware that cannot meet computing and power requirements of innovative functions [18], [19]. Computation offloading addresses this by dynamically distributing and executing functions within communication networks rather than in vehicles, reducing resource requirements while improving quality of experience and service [20], [21]. This approach utilizes cloud data centers and edge nodes for edge intelligence [22]. While cloud data centers offer scalability and power with high latency limitations, edge nodes provide lower latency for real-time applications despite reduced power [23]. The offloading process involves five steps: function decomposition, dynamic target node selection, resource allocation, relocation processes, and result reporting [18], [24].

C. Data Loop The rapid update cadence and complex interplay of advanced, connected functions in SDVs create a dynamic IT infrastructure requiring sophisticated management approaches. Traditional testing approaches are insufficient for managing frequent updates in complex systems. CI/CD practices provide automated testing frameworks, but system complexity renders exhaustive testing impractical. Instead, targeted testing strategies focusing on risk areas are being researched [25], prioritizing tests based on their likelihood of being affected by updates to balance thoroughness and efficiency.

As software bugs increasingly reach operation, runtime diagnostics become paramount. Static diagnosis techniques, traditional in automotive systems, are becoming obsolete. The service-oriented vehicle diagnostics (SOVD) standard offers a flexible approach to diagnosing vehicle systems, providing adaptable frameworks for SDVs' dynamic nature. Advanced diagnosis techniques consider dependencies and data flows between distributed components. By enhancing data lakes with machine learning algorithms and live data analytics, these approaches proactively identify and mitigate issues, ensuring optimal performance and reliability in evolving systems [26].

Modern vehicles perform advanced functions requiring vast data leverage through ML models to enhance user comfort and prevent safety risks [27]. Traditional centralized ML model training raises privacy concerns in the Big Data era [28]. Federated learning (FL) emerged as an alternative, training models directly on user devices to mitigate privacy risks by avoiding centralized raw user data access [29]. Despite privacy-preserving technologies such as k-Anonymity [30], l-Diversity [31], or t-Closeness [32], privacy breach risks persist. Recent studies demonstrate adversaries can reconstruct user-specific data from model updates using generative adversarial networks or inference attacks [33], [34], posing risks throughout the learning process, particularly in automotive domains with multiple stakeholders. Users hesitate to share local data even when stored in OEM clouds, creating safety risks through potential local user data reconstruction [35]. Privacy techniques include homomorphic encryption [36], differential privacy [37], and secure

Authorized licensed use limited to: Universidad de Malaga. Downloaded on September 13,2026 at 16:03:57 UTC from IEEE Xplore. Restrictions apply.

multiparty communication [38], each with distinct advantages and disadvantages requiring use case-specific assessment.

Increasing vehicle networking creates new attack vectors for cyber criminals [39], posing risks to privacy, personal safety, road traffic stability, and national security. In a survey conducted in 2014, 65% stated that they were at least moderately concerned about vehicle security [40]. Attack targets include data theft and access to driving functions like engine control, steering, and brakes. The interconnection between security and safety becomes increasingly evident, similar to Industry 4.0 applications. Vehicle communication can be disrupted through distributed denial of service attacks or signal jamming, making complete radio link dependency for safety imperative to avoid. Cyber-security procedures for vehicles should be standardized similarly to safety procedures in ISO 26262 [41].

D. Economy and Politics The economic and political dimensions of SDVs present both opportunities and challenges for traditional automotive business models and regulatory frameworks. Today, significant portions of vehicle software components are published under open source licenses, largely due to Linux-based operating systems proliferation in modern ECUs and HPCs [42].

Open source software accelerates development by pooling engineering efforts, making development more cost-effective. However, liability considerations in damage events and repository management questions must be addressed. Beyond open source considerations, the broader regulatory landscape presents significant challenges for SDV implementation. The remarkable technology transformation in the mobility space has outpaced regulatory frameworks, with governments operating under outdated paradigms [43].

Extending openness to both software and hardware interfaces represents a radical departure from current industry practice, shifting traditional linear supply chain dependence to flexible, value-creating networks [44]. This transformation enables new business models and partnerships in the automotive ecosystem, such as marketplaces for vehicle-related services [45]. Moreover, the utilization of artificial intelligence (AI) for a variety of functions within the vehicle offers a lot of potential. Start-ups are able to concentrate on a particular function and therefore develop a specific AI model for that one application [46]. This is an advantage over larger organizations that may lack the resources to develop several different AI functions.

Consumer expectations, shaped by personal electronics, now demand frequent feature enhancements and security patches. However, traditional automotive development processes are reaching their limits, with established cycles measuring in years due to complex component integration and extensive safety regulation compliance. As SDVs incorporate more software functionality, the fundamental tension between rapid, iterative updates and the methodical validation approaches necessary for automotive safety becomes increasingly complex, requiring new development methodologies that can accommodate both software agility and automotive rigor [47].

III. Methodology

In this section, we outline our methodology used to analyze the current trends and challenges of the SDV and the possible directions of future research questions. Our analysis is based on three key sources of information. These are:

(a) expert interviews with automotive IT specialists;
(b) study on major automotive trade fairs and workshops;
(c) authors' professional experience.

Through the analysis of these sources, we established the current trends in the SDV field. Consequently, we identified challenges associated with these trends that must be addressed to facilitate future advancements (see Figure 1).

The following gives an overview of our three sources of information.

A. Expert Interviews With Automotive IT Specialists

The primary source for identifying current trends and challenges in the field of SDVs is a survey of experts. To conduct this survey, a catalog of 32 questions was created and distributed among the contributors and partners of the "Software-Defined Car" (SofDCar) project (see Section III-C) in February 2024. A total of 27 individuals participated in the survey, with 48% of respondents being affiliated with research institutions such as the University of Stuttgart or the Karlsruhe Institute of Technology (KIT). Approximately 30% of the participants reported working for an automotive supplier, while the remaining 22% were from an original equipment manufacturer (OEM). This achieved a good cross section of participants, allowing for the derivation of certain trends from the survey.

B. Study on Major Automotive Trade Fairs and Workshops

Automotive trade fairs are not only an important place to present new products. They can also be used to identify industry trends and gaps in the market. For this reason, three important trade fairs were selected to capture the current trends in the field of SDVs.

Start-ups are especially interesting because these companies have to present new innovations in order to gain a foothold in the automotive

Authorized licensed use limited to: Universidad de Malaga. Downloaded on September 13,2026 at 16:03:57 UTC from IEEE Xplore. Restrictions apply.

sector. Many such start-ups are headquartered in Israel. Therefore, the "EcoMotion Week" took place in May 2023. Around 100 exhibitors were present at this trade fair, and a further 33 were presented in an additional catalog. Based on the exhibitor descriptions, 99 companies working in the field of SDV were analyzed.

In order to compare the "EcoMotion Week" with other international trade fairs, we decided to also analyze the exhibitors at the German "International Automobile Exhibition" (IAA) and the American "Consumer Electronics Show" (CES).

In September 2023, the IAA took place in Munich. A total of 671 companies are listed in the exhibitor and partner directory. To ensure that only the relevant exhibitors are included in this analysis, only the category "automated and autonomous driving, driver assistance systems" was taken into account, which corresponds to 61 exhibitors.

According to the CES in Las Vegas in January 2024, there were over 4300 exhibitors. Of these, 444 companies were assigned to the "Vehicle Tech and Advanced Mobility" category on the CES website. A total of 137 of the 444 companies were included in our analysis. Companies without a description or exhibitors from the areas of batteries, motors, etc., were not included in order to consider only companies in the SDV sector.

The results of this trade fairs analysis were subsequently compared with the research portfolios of 98 funded research projects by the German Federal Ministry for Digital and Transport (BMDV), the Federal Ministry for Economic Affairs and Climate Action (BMWK), and the Federal Ministry of Education and Research (BMBF) with a view to identifying similarities and differences between the research priorities of industry and academia. The project profiles were published at a symposium in December 2023 in Berlin.

C. Authors' Professional Experience

The third source of information is the knowledge and experience the authors have gained from various research projects, workshops, and conferences. The most significant project is the SofDCar,[1] which is nationally funded by the German Federal Ministry for Economic Affairs and Climate Action. In total, around 200 engineers and researchers from different companies and universities participate in researching new methods and technologies that are essential for the emergence of the SDV.

In total, over the past five years, the authors took part in ten industry projects concerned with connected automotive systems (4 concluded, 5 running and 1 planned). One example is the recently launched EU-funded project "Hardware Abstraction Layer for the Software-Defined Vehicle" (HAL4SDV), which is tasked with the conceptualization of an EU-wide OpenSource solution for virtualized vehicle functions and connectivity across OEMs. As part of these projects and our own research, we have also participated in several workshops such as the "Standardized Automotive Software Workshop" in November 2023, where representatives from industry and research institutions discussed various aspects of future vehicles in depth.

[1] https://sofdcar.de/

IV. Findings

In this section, we present the results of our holistic study of current trends regarding the SDV. Our analysis resulted in 11 major trends, from which we derived a number of challenges that need to be addressed in the future to foster these trends.

The trends and challenges are grouped into the three dimensions which were outlined in Section II.

(a) Architecture.
(b) Data loop.
(c) Economy and politics.

A. Architecture

Trend 1: Toward a new digital ecosystem for the SDV.

For the realization of the SDV, it is imperative not only to make adjustments to the car itself

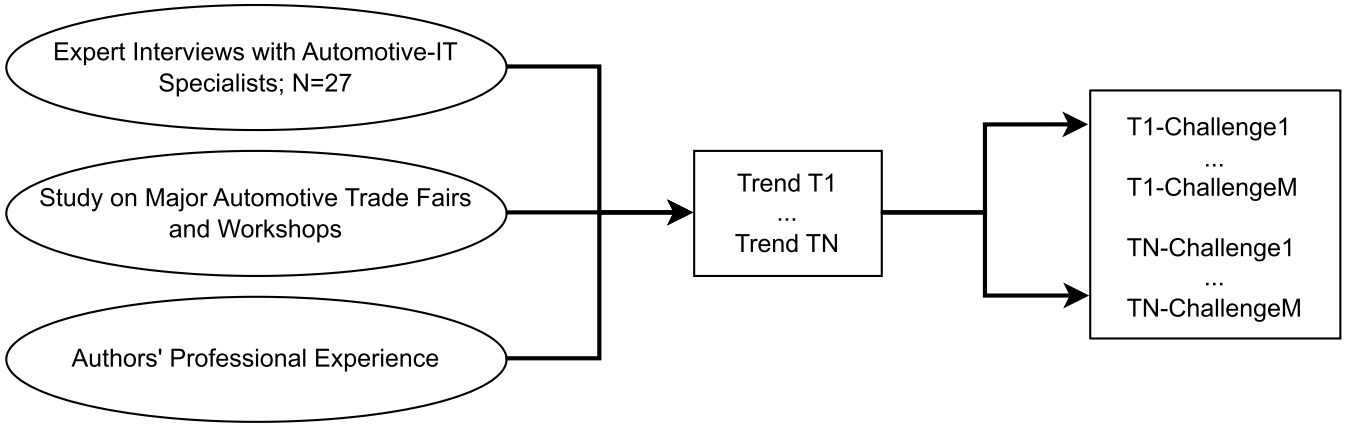


Figure 1. Research methodology to analyze trends and derive challenges.

Authorized licensed use limited to: Universidad de Malaga. Downloaded on September 13,2026 at 16:03:57 UTC from IEEE Xplore. Restrictions apply.

by renewing or reinventing software architectures or even E/E architectures, but the entire ecosystem surrounding the automobile must undergo a complete transformation. As it is displayed in Figure 2, the transition involves a novel digital ecosystem consisting of digital infrastructure, digital services, and the digital vehicle, that all play a vital role in the new digital SDV ecosystem. Due to the motivation behind SDV, aiming for seamless connectivity among all players in this vast system, this new ecosystem is inevitable. Within this ecosystem, the digital infrastructure encompasses both edge and cloud architectures, the significance of which is underscored by our interview findings, highlighting their strategic importance (see Figure 5). This new infrastructure offers performance enhancement, real-time data analysis, improved security, optimized resource utilization, and flexibility for adapting to changing requirements throughout the lifecycle of the SDV. It enables faster processing of data locally in the vehicle, while leveraging the cloud for scalable computing tasks, enhancing overall efficiency and adaptability. Critical to the infrastructure is the question of ownership and responsibility: who provides, finances, maintains, and governs it? Is it the government, OEMs, or tech companies? This question becomes particularly pertinent when considering the massive infrastructure investments required and the regulatory implications of such a distributed system.

Moving to digital services, a plethora of connected mobility offerings emerges, including ride-hailing, remote vehicle diagnostics and maintenance, personalized entertainment, and information services during travel, alongside seamless integration with smart home systems. Furthermore, digital services encompass the concept of Smart Cities, incorporating traffic management and optimization strategies, such as real-time traffic flow monitoring, congestion prediction, and parking space management utilizing smart sensors to reduce search time and optimize space utilization. These services represent a fundamental shift from vehicles as isolated transportation units to integral components of a connected urban infrastructure. Finally, within this ecosystem, the digital vehicle represents a significant focus, as outlined in the following.

We have derived the following challenges from this:

T1-Challenge1: Managing the interaction between the digital infrastructure, the digital services as well as the digital vehicle in the new ecosystem.

T1-Challenge2: Realizing the new ecosystem and answering the questions regarding the provision, maintenance and governance of the new ecosystem by assigning roles.

Trend 2: Zonal architecture with HPCs as the future architecture of the SDV.

The evolution from traditional distributed ECU architectures to consolidated, zonal approaches represents one of the most significant structural changes in automotive engineering. Our empirical findings

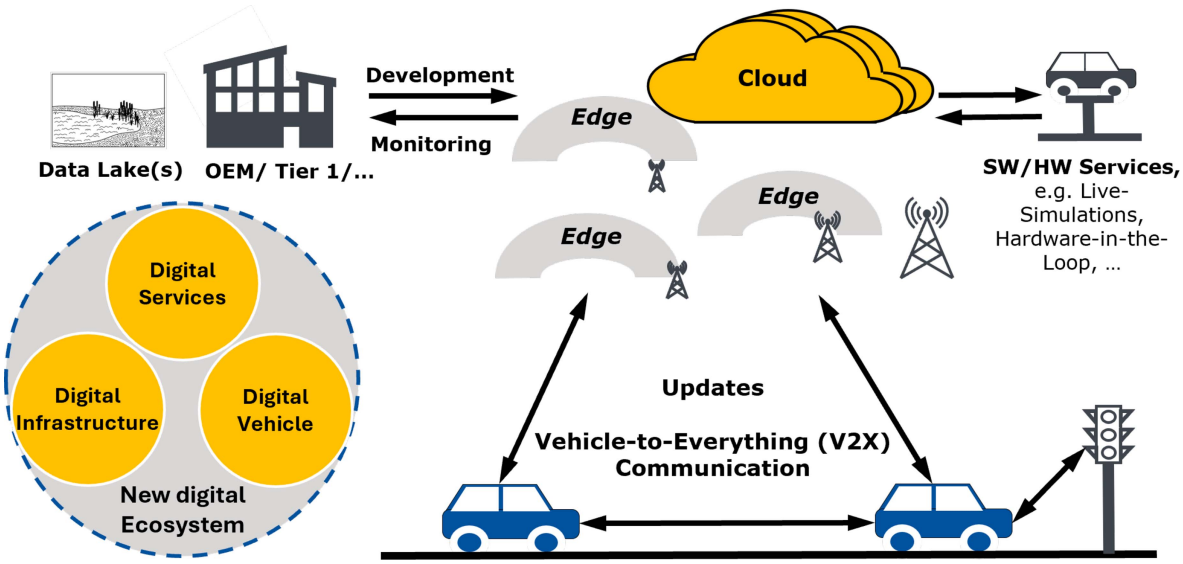


Figure 2. New digital ecosystem for the SDV.

Authorized licensed use limited to: Universidad de Malaga. Downloaded on September 13,2026 at 16:03:57 UTC from IEEE Xplore. Restrictions apply.

confirm that HPCs have emerged as the critical enabling technology for SDV implementation, directly addressing the computational limitations that constrain advanced vehicle functions. The architectural consolidation enables the abstraction layers and unified approaches necessary for true software-defined functionality. This transition aligns with the new concept of E/E architecture as defined in the European Commission's concept paper, and notably, all projects we participated in relied on this architectural paradigm, demonstrating its practical adoption across the automotive industry. This trend toward HPCs as the main computing units in the vehicle is consistent with the results of our interview. As shown in Figure 3, HPCs were recognized as a relevant technology for the implementation of the SDV. The migration to zonal architectures with HPCs represents not merely a technical upgrade, but a foundational prerequisite for enabling advanced features such as autonomous driving, real-time data processing, and dynamic function allocation that define the SDV.

We have derived the following challenges from this:

T2-Challenge1: Integration and coordination of various functions and applications within the zones.

T2-Challenge2: Transfer of the existing ECU software to HPCs.

Trend 3: Mastering the high variance in SDVs.

The increasing complexity of automotive software systems has created unprecedented challenges in managing variability across multiple dimensions. High configurability requirements combined with continuous evolution through rapid development cycles generate a complexity matrix that traditional development approaches cannot effectively handle. Our research findings validate that managing this variability has become a decisive factor for SDV realization, with systematic approaches to software reuse and product line engineering emerging as essential methodologies. The results of our interview underline the relevance of this complexity problem and therefore the trend to find suitable solutions to master the variability in SDVs. As depicted in Figure 4, two thirds of respondents see the mastering of variability as a clearly decisive factor for the realization of the SDV. Moreover, the prominence of the Software Product Line Conference as the second most prolific venue for publications in automotive software engineering, alongside the VaMos (Variability Modelling of Software-Intensive Systems) workshop ranking within the top 5 most publication-intensive workshops, underscores the criticality of managing complexity stemming from high variability in vehicles [48]. In addition, the recognition of Software Reuse as one of the top 3 research topics in automotive software engineering, alongside System/Software Architecture and Design, further accentuates the imperative to address this complexity in the automotive domain [48].

We have derived the following challenges from this:

T3-Challenge1: Evolving a high variance of software configurations over a longer period of time.

T3-Challenge2: Automating the complex task of developing and managing variance-rich automotive software.

Trend 4: Function off-loading as a future technology.

As vehicle functions become increasingly sophisticated, traditional on-board computing resources face fundamental limitations in meeting performance and power requirements. Computation offloading emerges as a critical enabling technology that allows dynamic distribution of functions across network infrastructure. Our empirical research strongly validates this direction, with overwhelming industry consensus recognizing cloud and edge computing as fundamental to SDV success. The balance between cloud scalability and edge proximity for real-time applications, as described in the literature, emerges as a critical

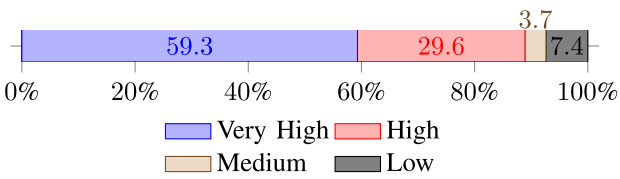


Figure 3. Importance of HPC architectures for the SDV.

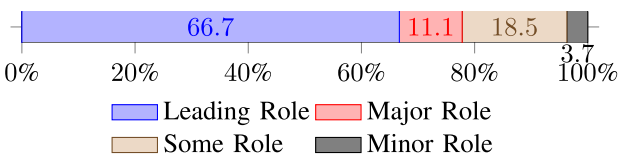


Figure 4. Role of mastering variability in SDVs.

Authorized licensed use limited to: Universidad de Malaga. Downloaded on September 13,2026 at 16:03:57 UTC from IEEE Xplore. Restrictions apply.

architectural decision that will define SDV performance characteristics. In the conducted interview (see Figure 5), 89% said the cloud will play an important or very important role in the SDV in the future. When it comes to processing at the edge, 81% expect it to play an important or very important role. This consensus reflects the industry's understanding that function offloading is not merely an optimization opportunity, but a fundamental requirement for enabling the advanced capabilities that distinguish SDVs from traditional vehicles. The convergence of high industry expectations with the technical necessity outlined in literature suggests that offloading capabilities will become a competitive differentiator in the SDV market.

We have derived the following challenges from this:

T4-Challenge1: Various functions exist with specific requirements regarding latency, bandwidth, and reliability. These requirements are requested by vehicles with different system configurations.

T4-Challenge2: Maintaining short execution time while keeping the error rate during the function execution low at the same time.

T4-Challenge3: Detecting available computation nodes within the network for offloading and their free computation capacity.

B. Data Loop

Trend 5: Automated testing and diagnosis strategies to handle complexity.

The accelerated software evolution and complex interdependencies characteristic of advanced vehicle systems present validation challenges that surpass conventional automotive testing paradigms. Thus, automated testing and diagnostic capabilities are becoming fundamental prerequisites for SDV viability. The results of our interview confirm the relevance of this topic to the field of the SDV. As Figure 6 shows, 85.2% of all participants in the survey deem the importance of automated testing and diagnosis strategies to be at least important, while 0% answered the importance to be low or very low. A similar picture emerges from the analysis of the funded research projects. Here, over 17% of projects focus on data collection and processing, the second highest share after road safety (see Figure 7). The importance of this topic is also recognized by the SofDCar project, where testing and diagnosis play the major role in the data loop work-stream (one out of four project parts). Moreover, the InnovationCampus Future Mobility (ICM) project TESSOF[2] research works how standardized and efficient test methods can be used for highly configurable vehicular software systems. These research initiatives demonstrate the practical urgency of developing sophisticated testing methodologies that can handle the complexity and variability inherent in SDV systems.

We have derived the following challenges from this:

T5-Challenge1: Testing software updates thoroughly while maintaining

[2] https://www.icm-bw.de/en/projects/project-overview/details/sdmobi5-tessof

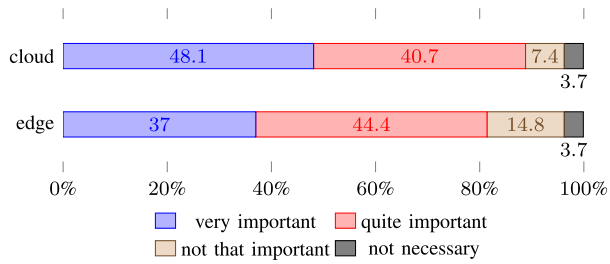


Figure 5. Importance of edge and cloud computing.

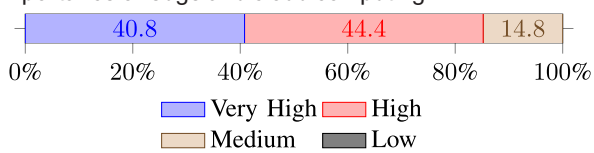


Figure 6. Future importance of automated test and diagnosis strategies for SDVs.

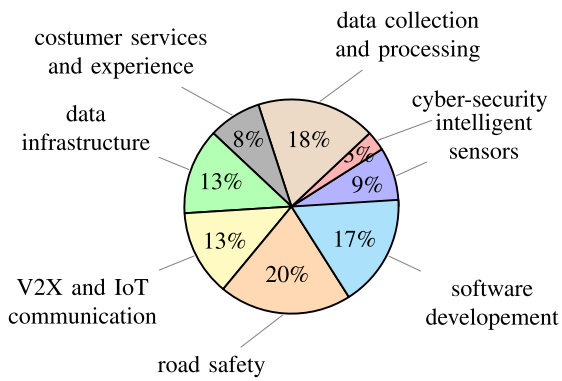


Figure 7. Focuses of the funded research projects in Germany.

Authorized licensed use limited to: Universidad de Malaga. Downloaded on September 13,2026 at 16:03:57 UTC from IEEE Xplore. Restrictions apply.

frequent deployments.

T5-Challenge2: Understanding dependencies and data flows between flexible, distributed components.

T5-Challenge3: Automating software diagnosis to proactively identify and mitigate issues.

Trend 6: Toward more central data collection and compromising privacy.

The sophisticated ML models that enable advanced SDV capabilities create an inherent tension between data requirements and privacy preservation. While centralized data collection offers the most efficient path to training powerful AI systems that enhance user comfort and safety, this approach directly conflicts with user privacy expectations and regulatory requirements. Our research reveals that this challenge has moved beyond theoretical concern to become a practical barrier to SDV adoption and capability advancement. The results of our survey indicate that the majority of respondents perceive a significant potential for such intelligent cockpits (see Figure 8). Subsequently, personalization can be integrated into this learning process to meet individual requirements, ensuring the model's performance with strong generalization capabilities and personalized parameters. This approach not only mitigates privacy concerns in such tasks but also enhances automated personalization to accommodate individual preferences effectively. However, users are hesitant to share their local data, such as personal preferences and location information, even when it is stored in the OEM cloud, as given in Figure 9.

We have derived the following challenges from this:

T6-Challenge1: Preventing the disclosure of sensitive information in learning tasks.

T6-Challenge2: Providing model updates in a secure manner without disrupting the user experience.

T6-Challenge3: Lack of automatic personalization of functions based on advanced approaches.

Trend 7: Cyber-security is becoming one of the most important technologies in the SDV.

As vehicles become integral components of digital infrastructure, cyber-security evolves from a secondary consideration to a primary safety requirement. However, a significant gap exists between industry recognition of cyber-security importance and actual implementation efforts.

The interview revealed that cyber-security was perceived as the most important technology for the SDV, with over 80% of respondents indicating this view, as seen in Figure 10. Our review of trade fair evaluations indicates that only few companies address security risks (see Figure 11). At major international trade fairs, such as IAA and CES, the proportion of exhibitors focusing on vehicle cyber-security is notably low. In addition, a significant number of German-funded research projects rarely prioritize security. This situation requires urgent attention, as road users should not only be protected when accidents have already occurred due to a lack of security. The disconnect between stated importance and actual investment suggests systemic challenges in translating security awareness into concrete protective measures.

We have derived the following challenges from this:

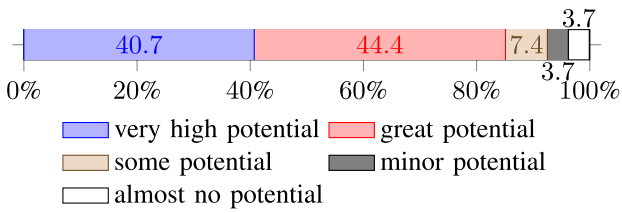


Figure 8. Potential of integrating smart devices into an intelligent cockpit.

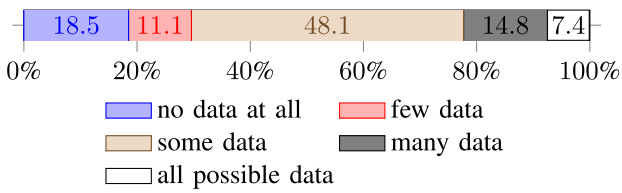


Figure 9. To what extent are you willing to share your personal data collected in the vehicle with OEMs?.

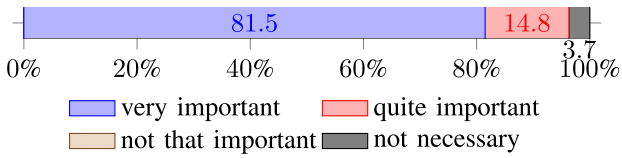


Figure 10. Importance of cyber-security in the SDV.

Authorized licensed use limited to: Universidad de Malaga. Downloaded on September 13,2026 at 16:03:57 UTC from IEEE Xplore. Restrictions apply.

T7-Challenge1: It is to be expected that cyber-attacks against vehicles will become an increasingly prevalent threat in the future.

T7-Challenge2: A standardized procedure to increase cyber-security should be established in a manner similar to the standardized procedure to ensure safety.

T7-Challenge3: At present, there is a deficiency in the investment of resources into the field of cyber-security in relation to vehicles.

C. Economy and Politics

Trend 8: Divergence Between Industry and Research Priorities.

The exhibitors present at the analyzed trade fairs (ECOM in 2023, IAA in 2023 and CES in 2024) were divided into different categories.

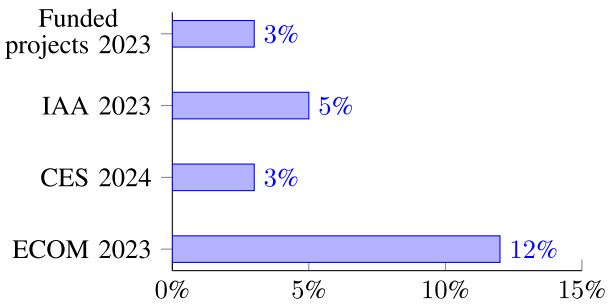


Figure 11. Percentage of exhibitors and research projects focusing on cyber-security in vehicles.

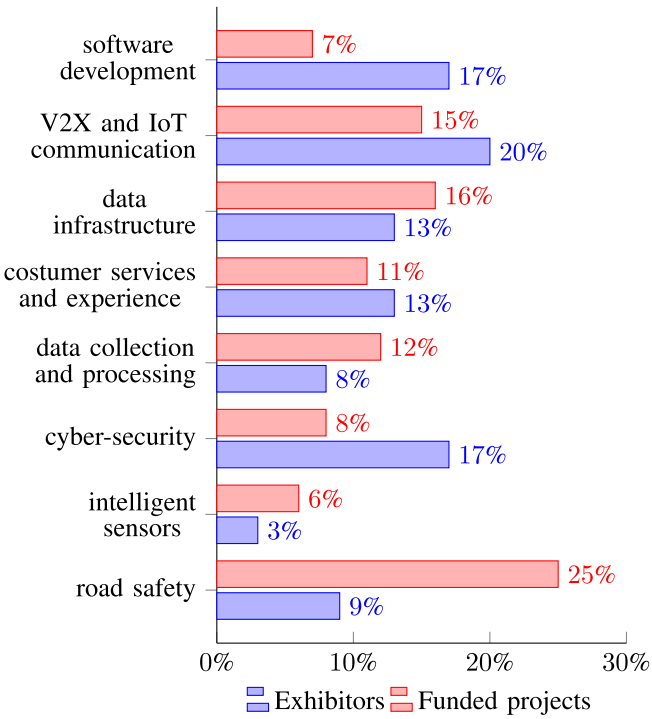


Figure 12. Comparison of exhibitors at the trade fairs with the funded projects.

These categories were determined by clustering the various similar exhibitors. In particular, the exhibitors present were divided into the following eight categories: Intelligent Sensors, cyber-security, Data Collection and Processing, Costumer Services and Experience, Data Infrastructure, V2X and IoT, Road Safety, and Software Development.

In a second step, all exhibitors were checked again to see whether they also fit into a second or sometimes even a third category. The most important factor for the categorization was the description of the exhibitors in the trade fair brochure.

As illustrated in Figure 12, there is a notable disparity between the proportion of exhibitors and the proportion of research projects that can be classified within a given category in some instances.

In the majority of cases, companies focus their research efforts on a specific subject area when there is a current need for it. In contrast, funded research should address topics that are relevant both now and in the future. It seems that intelligent sensors, which are mostly equipped with AI-supported evaluation, have become a prevalent development trend for new vehicle models. However, at the same time, there seems to be a lack of dedicated research focus on this topic, as these technologies have already made their way into the industry and it seems unlikely that any significant breakthroughs will be made in this area.

In contrast, data processing and software development are subjects of greater interest to researchers than they are to trade fair exhibitors. The development of new methods based on so-called “large language models” (LLMs), which are currently being tested in a number of new areas, is likely to be a factor here [49].

Authorized licensed use limited to: Universidad de Malaga. Downloaded on September 13,2026 at 16:03:57 UTC from IEEE Xplore. Restrictions apply.

The most crucial aspect of research is maintaining focus. While it is essential to evaluate novel methodologies to advance technological progress, these approaches must be application-oriented to some extent. Otherwise, research outcomes may be swiftly superseded by other technologies.

At the same time, the industry must be cautious to monitor emerging research trends. Research is concerned with identifying solutions to problems that are not always directly related to the primary customer needs. However, these are often not to be overlooked when it comes to safety or the automation and increased efficiency of processes.

We have derived the following challenges from this:

T8-Challenge1: Research in the field of SDV must not become too abstract, but must remain application-orientated.

T8-Challenge2: The industry must also adapt research findings that do not immediately serve to increase sales.

Trend 9: Open source software and open interfaces in SDVs.

The proliferation of open source software in automotive systems creates both opportunities and challenges for industry transformation. While openness can accelerate development and foster innovation through collaborative approaches, it also introduces complex questions about liability, governance, and competitive strategy in safety-critical applications. Our empirical findings reveal fundamental tensions between the theoretical benefits of open development and the practical constraints imposed by competitive realities and regulatory requirements in the automotive domain. The literature's discussion of liability concerns, maintenance responsibilities, and governance challenges finds direct practical relevance in our findings, particularly regarding the contradiction between OEM desires for open infotainment systems and reluctance to share data. This paradox highlights the gap between the theoretical benefits of openness outlined in academic literature and the practical constraints imposed by competitive and regulatory realities. This view is shared by the majority of respondents to our survey, as shown in Figure 13. The challenge for OEMs is that while they would prefer an open infotainment system, they are also reluctant to share data. This was demonstrated at the SAS workshop in November 2023, where experts from research and industry engaged in intensive discussions. The workshop revealed fundamental tensions between the theoretical advantages of open systems and the practical concerns about competitive advantage, data protection, and liability that constrain implementation. The emergence of licensed providers as intermediaries, as mentioned in the literature, represents one approach to resolving these tensions, but the broader questions of governance, standardization, and long-term sustainability of open automotive software ecosystems remain largely unresolved in practice.

We have derived the following challenges from this:

T9-Challenge1: OEMs must disclose software and hardware interfaces without publishing secret data.

T9-Challenge2: The question of liability in the event of an error in open source software must be clarified.

T9-Challenge3: Open source software must be further developed and maintained in the long term.

T9-Challenge4: The administration of open source software must be coordinated by an institution; it is unclear who this institution should be.

Trend 10: A growing number of new players will contribute to the future advancement of the SDV.

As the proportion of software in vehicles increases, so does the number of software suppliers. As the evaluation of the trade fairs showed (see Section II-B), there are many new start-ups in this area, offering various software solutions for sensors or actuators, often in conjunction with AI. It can therefore be expected that the share of start-ups in the development of a vehicle will increase if the OEMs do not counteract this trend. Concurrently, established competitors in the IT and infotainment sector will enter the automotive OEM market. German car manufacturers and US IT companies are already

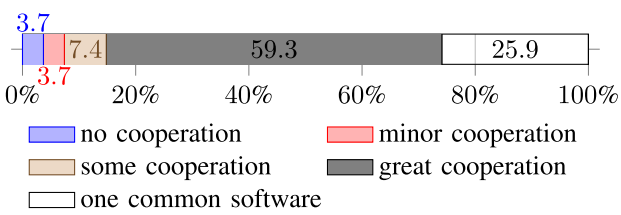


Figure 13. Need for collaboration with respect to software and open interfaces.

Authorized licensed use limited to: Universidad de Malaga. Downloaded on September 13,2026 at 16:03:57 UTC from IEEE Xplore. Restrictions apply.

cooperating to run operating systems in cars that are familiar to customers. This raises various questions for OEMs regarding their competitive strategy. These include which operating system should be chosen for which vehicle model, who can or may develop applications for the infotainment system, and what authorizations should be granted to external software. In order to prevent a reduction in their share of the value chain of a vehicle in the future, OEMs and their traditional suppliers must invest more in the development of software and expand their knowledge in AI. Conversely, the use of open interfaces (see trend 9) may also confer a competitive advantage, as it allows other providers to offer their software products in the car.

We have derived the following challenges from this:

- T10-Challenge1: The advent of new software providers has intensified the competitive pressure on established suppliers and OEMs.
- T10-Challenge2: In order to remain competitive, it is essential to invest further in software and AI.
- T10-Challenge3: A delicate balance must be struck between in-house software and external applications in the infotainment sector.

Trend 11: Modern development cycles change the process of integration and delivery.

The automotive industry faces the challenge of meeting consumer expectations for frequent software updates while maintaining rigorous safety validation processes, as traditional multiyear development cycles clash with the demand for rapid, continuous feature enhancements and security patches. At the same time, the growing number of software suppliers (trend 10) and the adoption of open software—and even hardware—interfaces increase both the frequency and complexity of such updates. As a result, integration and test methodologies need to evolve toward more modular processes that can accommodate frequent changes while maintaining the stringent quality assurance essential for safety-critical automotive environments.

Building on these challenges, the newly accepted Dev4Test project addresses the need for a fundamental shift towards modern test-centered development processes. Thus, placing quality assurance at the core of the development process from the very beginning, ensuring that testing is not a separate downstream activity, but an integral, continuous driver of development. Here, a trend toward modular, iterative processes that can manage growing system complexity and variability without compromising safety is visible. Testing is integrated seamlessly across all stages—from simulation to Hardware-in-the-Loop to real-world vehicle validation—enabling early detection of issues and faster adaptation to change. These modern processes also emphasize transparency and traceability, which are essential for certification and regulatory approval. While data-driven and AI-supported methods are being explored for tasks like test optimization or fault prediction, the focus remains on ensuring their results are explainable and reliable. This evolution in development methodology supports higher productivity and flexibility, helping industry players keep pace with the rapid innovation cycles expected in the era of software-defined vehicles.

We have derived the following challenges from this:

- T11-Challenge1: Design systems, their subsystems, and interfaces in a way that asynchronous development and orchestration becomes possible.
- T11-Challenge2: Adapt automotive development processes to support significantly shorter integration and delivery cycles.
- T11-Challenge3: Ensure Verification and Validation throughout the development process despite the increasing complexity.

V. Discussion

The analysis of current trends in the SDV domain reveals a dynamic and evolving landscape with far-reaching implications for industry, academia, and society at large. This discussion chapter aims to delve deeper into the complexities and interdependencies of these trends, exploring their implications and opportunities.

We summarized our results of the study in Table 1. These results clearly state that the idea of a SDV encompasses much more than just the vehicle itself and its functionality. Rather, an entire new ecosystem must be considered here, which is still in the early stages of its development and brings with it a number of unanswered questions, such as the ownership of this ecosystem. In addition, it is essential to acknowledge the multitude of regulations that either impede or are yet to be established.

Authorized licensed use limited to: Universidad de Malaga. Downloaded on September 13,2026 at 16:03:57 UTC from IEEE Xplore. Restrictions apply.

These aspects are entirely separate from the vehicle's technology but pose significant hurdles in the SDV development, requiring resolution to advance progress. This includes open interfaces, open-source software, and defined roles in ecosystem construction and management. As SDVs become increasingly integrated into everyday life also, ethical and social considerations surrounding privacy, data security, and equity become paramount. Striking a balance between technological advancement and ethical responsibility requires proactive measures, such as implementing privacy-preserving

Table 1. Summary of Analyzed Trends and Derived Challenges.

Trends		**Challenges**	
Trend 1:	Towards a new Digital Ecosystem for the SDV.	Challenge 1:	Managing the interaction between the digital infrastructure, the digital services, as well as the digital vehicle in the new ecosystem.
		Challenge 2:	Realizing the new ecosystem and answering the questions regarding the provision, maintenance and governance of the new ecosystem by assigning roles.
Trend 2:	High Performance Computing Architectures are the future architecture of the SDV.	Challenge 1:	Integration and coordination of various functions and applications within the zones.
		Challenge 2:	Transfer of the existing ECU software to HPCs.
Trend 3:	Mastering the high variance in SDVs.	Challenge 1:	Evolving a high variance of software configurations over a longer period of time.
		Challenge 2:	Automating the complex task of developing and managing variance-rich automotive software.
Trend 4:	Function offloading as a future technology.	Challenge 1:	Various functions exist with specific requirements regarding latency, bandwidth and reliability. These requirements are requested by vehicles with different system configurations.
		Challenge 2:	Maintaining short execution time while keeping the error rate during the function execution low at the same time.
		Challenge 3:	Detecting available computation nodes within the network for offloading and their free computation capacity.
Trend 5:	Automated testing and diagnosis strategies to handle complexity.	Challenge 1:	Testing software updates thoroughly while maintaining frequent deployments.
		Challenge 2:	Understanding dependencies and data flows between flexible, distributed components.
		Challenge 3:	Automating software diagnosis to proactively identify and mitigate issues.
Trend 6:	Towards more central data collection and compromising privacy.	Challenge 1:	Preventing the disclosure of sensitive information in learning tasks.
		Challenge 2:	Providing model updates in a secure manner without disrupting the user experience.
		Challenge 3:	Lack of automatic personalization of functions based on advanced approaches.
Trend 7:	Cyber-security is becoming one of the most important technologies in the SDV.	Challenge 1:	It is to be expected that cyber-attacks against vehicles will become an increasingly prevalent threat in the future.
		Challenge 2:	A standardized procedure for increasing cyber-security should be established in a manner similar to the standardized procedure for ensuring safety.
		Challenge 3:	At present, there is a deficiency in the investment of resources into the field of cyber-security in relation to vehicles.
Trend 8:	Divergence Between Industry and Research Priorities.	Challenge 1:	Research in the field of SDV must not become too abstract, but must remain application-orientated.
		Challenge 2:	The industry must also adapt research findings that do not immediately serve to increase sales.
Trend 9:	Open source software and open interfaces in SDVs.	Challenge 1:	OEMs must disclose software and hardware interfaces without publishing secret data.
		Challenge 2:	The question of liability in the event of an error in open source software must be clarified.
		Challenge 3:	Open source software must be further developed and maintained in the long term.
		Challenge 4:	The administration of open source software must be coordinated by an institution; it is unclear who this institution should be.

Authorized licensed use limited to: Universidad de Malaga. Downloaded on September 13,2026 at 16:03:57 UTC from IEEE Xplore. Restrictions apply.

Table 1. (Continued).

Trends		Challenges	
Trend 10:	A growing number of new players will contribute to the future advancement of the SDV.	Challenge 1:	The advent of new software providers has intensified the competitive pressure on established suppliers and OEMs.
		Challenge 2:	In order to remain competitive, it is essential to invest further in software and AI.
		Challenge 3:	A delicate balance must be struck between in-house software and external applications in the infotainment sector.
Trend 11:	Modern development cycles change the process of integration and delivery.	Challenge 1:	Design systems, their subsystems and interfaces in a way that asynchronous development and orchestration becomes possible.
		Challenge 2:	Adapt automotive development processes to support significantly shorter integration and delivery cycles.
		Challenge 3:	Ensure Verification and Validation throughout the development process despite the increasing complexity.

data collection mechanisms and ensuring transparency in decision-making algorithms. The development and deployment of SDVs necessitates robust regulatory and policy frameworks to ensure safety, security, and compliance with legal requirements. Standardization efforts, certification processes, and government regulations play a crucial role in establishing a conducive environment for innovation while safeguarding public interests.

In regard of the technology trends including OTA updates, mastering the high software variability and the growing importance of high-performance computing architectures, the derived trends are interconnected and mutually reinforcing. For instance, the adoption of agile OTA updates necessitates robust cyber-security measures and a consistent variant management, while advancements in high-performance computing architectures enable the realization of innovative SDV functionalities. The integration of emerging technologies, such as function offloading, automated testing and diagnosis strategies, holds promise for further enhancing the capabilities of OTA updates.

The differing priorities between industry stakeholders and academic researchers highlight the need for collaboration and knowledge exchange to address emerging challenges effectively. Industry practitioners are focused on meeting consumer demands for advanced features and functionalities, whereas researchers are exploring novel technologies and methodologies to address technical challenges and societal concerns. In conclusion, the analysis of trends in the SDV domain underscores the need for holistic and interdisciplinary approaches to address the multifaceted challenges and opportunities presented by technological advancements. By fostering collaboration, embracing ethical principles, and adopting a forward-thinking mindset, stakeholders can navigate the complexities of the SDV landscape and pave the way for a safer, more efficient, and equitable future of mobility.

VI. Conclusion and Future Work

In this work, we presented a comprehensive study on the current trends and challenges in the field of the SDV by focusing on an industrial perspective. We outlined our research method which included three sources of information from which we analyzed current trends in the industry of the SDV. We then derived challenges based on the analyzed trends to give an outlook of what to expect in the coming years. The results of our study reveal 11 trends in the categories architecture, data loop and economy and politics and thus span a complete view on the SDV.

In our future work, we intend to develop an intelligent infrastructure consisting of an intelligent backend, intelligent fleet, and intelligent cockpit. The intelligent backend will function as the central unit responsible for overseeing the update process, transmitting relevant information to both fleet components and vehicles to enhance functionalities. It will maintain regular communication with both the fleet and the cockpit. The intelligent fleet, on the other hand, will comprise a fleet of Autonomous Ground Vehicles equipped with multiple environmental sensing devices. These vehicles will maintain continuous communication with the intelligent backend, enabling them to perform driving scenarios and receive updates as needed, such as for collision avoidance or making crossover decisions. Meanwhile, the intelligent cockpit will be equipped with a variety of sensors to gather data on environmental parameters, physiological signals, and personal preferences. These data will be utilized to enhance comfort conditions across various scenarios through the training of ML models with distributed data.

Authorized licensed use limited to: Universidad de Malaga. Downloaded on September 13,2026 at 16:03:57 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] H. Vdovic, J. Babic, and V. Podobnik, "Automotive software in connected and autonomous electric vehicles: A review," *IEEE Access*, vol. 7, pp. 166365–166379, 2019.

[2] Z. Liu, W. Zhang, and F. Zhao, "Impact, challenges and prospect of software-defined vehicles," *Automot. Innov.*, vol. 5, no. 2, pp. 180–194, 2022, doi: 10.1007/s42154-022-00179-z#citeas.

[3] S. S. Musa, M. Zennaro, M. Libsie, and E. Pietrosemoli, "Convergence of information-centric networks and edge intelligence for IoV: Challenges and future directions," *Future Internet*, vol. 14, no. 7, 2022, Art. no. 192.

[4] D. F. Blanco, F. Le Mouël, T. Lin, and M.-P. Escudié, "A comprehensive survey on software as a service (SaaS) transformation for the automotive systems," *IEEE Access*, vol. 11, pp. 73688–73753, 2023.

[5] "Software-defined vehicles–a forthcoming industrial evolution," [Online]. Available: https://www2.deloitte.com/cn/en/pages/consumer-business/articles/software-defined-cars-industrial-revolution-on-the-arrow.html

[6] W. Xia, Y. Wen, C. H. Foh, D. Niyato, and H. Xie, "A survey on software-defined networking," *IEEE Commun. Surv. Tut.*, vol. 17, no. 1, pp. 27–51, Firstquarter 2015.

[7] S. Liu, "The business case for infrastructure-vehicle cooperative autonomous driving," *IEEE Eng. Manag. Rev.*, vol. 50, no. 2, pp. 189–194, Secondquarter 2022.

[8] S. Singh, V. More, and R. Batheri, "Driving electric vehicles into the future with battery management systems," *IEEE Eng. Manag. Rev.*, vol. 50, no. 3, pp. 157–161, Thirdquarter 2022.

[9] "Concept paper on an open European software-defined vehicle platform shaping Europe's digital future," Feb. 2023. [Online]. Available: https://digital-strategy.ec.europa.eu/en/library/concept-paper-open-european-software-defined-vehicle-platform

[10] "How can OEMs distinguish themselves through automotive software? McKinsey," [Online]. Available: https://www.mckinsey.com/industries/automotive-and-assembly/our-insights/rewiring-car-electronics-and-software-architecture-for-the-roaring-2020s

[11] L. Holsten, C. Frank, J. Krüger, and T. Leich, "Electrics/electronics platforms in the automotive industry: Challenges and directions for variant-rich systems engineering," in *Proc. 17th Int. Work. Conf. Variability Modelling Softw. -Intensive Syst.*, 2023, pp. 50–59.

[12] J. Stümpfle, N. Jazdi, and M. Weyrich, "Tackling erosion in variant-rich software systems: Challenges and approaches," *Procedia CIRP*, vol. 128, pp. 633–637, 2024.

[13] M. Eggert, K. Günther, J. Maletschek, A. Maxiniuc, and A. Mann-Wahrenberg, "In three steps to software product lines: A practical example from the automotive industry," in *Proc. 26th ACM Int. Syst. Softw. Product Line Conf. -Volume*, 2022, pp. 170–177.

[14] K. Ignaim and J. M. Fernandes, "An industrial case study for adopting software product lines in automotive industry: An evolution-based approach for software product lines (EVOA-SPL)," in *Proc. 23rd Int. Syst. Softw. Product Line Conf. -Volume*, 2019, pp. 183–190.

[15] M. Marques, J. Simmonds, P. O. Rossel, and M. C. Bastarrica, "Software product line evolution: A systematic literature review," *Inf. Softw. Technol.*, vol. 105, pp. 190–208, 2019.

[16] J. Stümpfle, S. Baum, D. Dittler, N. Jazdi, and M. Weyrich, "Automating software product line adoption based on feature models using large language models," in *Proc. IEEE 29th Int. Conf. Emerg. Technol. Factory Automat.*, 2024, pp. 1–4.

Authorized licensed use limited to: Universidad de Malaga. Downloaded on September 13,2026 at 16:03:57 UTC from IEEE Xplore. Restrictions apply.

[17] J. Stümpfle, D. Atray, N. Jazdi, and M. Weyrich, "Large language model assisted transformation of software variants into a software product line," in *Proc. IEEE/ACM 22nd Int. Conf. Softw. Syst. Reuse*, 2025, pp. 12–20.
[18] C. Feng, P. Han, X. Zhang, B. Yang, Y. Liu, and L. Guo, "Computation offloading in mobile edge computing networks: A survey," *J. Netw. Comput. Appl.*, vol. 202, 2022, Art. no. 103366.
[19] F. Dettinger et al., "Machine-learning-based fault detection in electric vehicle powertrains using a digital twin," SAE Tech. Paper, Tech. Rep., 2023, doi: 10.4271/2023-01-1214.
[20] D. Baumann, M. Sommer, F. Dettinger, T. Roesch, E. Sax, and M. Weyrich, "Connected vehicle: Ontology, taxonomy and use cases," in *Proc. IEEE Int. Syst. Conf.*, 2024, pp. 1–6, in press.
[21] B. C. Gül, N. Devarakonda, D. Dittler, N. Jazdi, and M. Weyrich, "Using federated learning in the context of software-defined mobility systems for predictive quality of service," 2023. https://www.vdi-nachrichten.com/shop/automation-2023/
[22] J. Liu et al., "RL/DRL meets vehicular task offloading using edge and vehicular cloudlet: A survey," *IEEE Internet Things J.*, vol. 9, no. 11, pp. 8315–8338, Jun. 2022.
[23] E. Mustafa et al., "Deep neural networks meet computation offloading in mobile edge networks: Applications, taxonomy, and open issues," *J. Netw. Comput. Appl.*, vol. 226, 2024, Art. no. 103886. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1084804524000638
[24] H. Lin, S. Zeadally, Z. Chen, H. Labiod, and L. Wang, "A survey on computation offloading modeling for edge computing," *J. Netw. Comput. Appl.*, vol. 169, 2020, Art. no. 102781.
[25] L. Güthing, M. Weiß, I. Schaefer, and M. Lochau, "Sampling cardinality-based feature models," in *Proc. 18th Int. Work. Conf. Variability Modelling Softw. -Intensive Syst.*, New York, NY, USA: Association for Computing Machinery, 2024, pp. 46–55.
[26] M. Weiß, M. Müller, F. Dettinger, N. Jazdi, and M. Weyrich, "Continuous analysis and optimization of vehicle software updates using the intelligent digital twin," in *Proc. IEEE 28th Int. Conf. Emerg. Technol. Factory Automat.*, 2023, pp. 1–7.
[27] B. C. Gül, N. Devarakonda, N. Jazdi, and M. Weyrich, "Personalized comfort features in software-defined vehicles using federated learning," in *Proc. IEEE 29th Int. Conf. Emerg. Technol. Factory Automat.*, 2024, pp. 1–7.
[28] K. Bonawitz et al., "Towards federated learning at scale: System design," *Proc. Mach. Learn. Syst.*, vol. 1, pp. 374–388, 2019.
[29] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. Artif. Intell. Statist.*, PMLR, 2017, pp. 1273–1282.
[30] L. Sweeney, "k-anonymity: A model for protecting privacy," *Int. J. Uncertainty, Fuzziness Knowl. -Based Syst.*, vol. 10, no. 05, pp. 557–570, 2002.
[31] A. Machanavajjhala, D. Kifer, J. Gehrke, and M. Venkitasubramaniam, "l-diversity: Privacy beyond k-anonymity," *ACM Trans. Knowl. Discov. From Data*, vol. 1, no. 1, pp. 3–es, 2007.
[32] N. Li, T. Li, and S. Venkatasubramanian, "T-closeness: Privacy beyond K-anonymity and l-diversity," in *Proc. IEEE 23rd Int. Conf. Data Eng.*, 2006, pp. 106–115.
[33] B. Hitaj, G. Ateniese, and F. Perez-Cruz, "Deep models under the GAN: Information leakage from collaborative deep learning," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, 2017, pp. 603–618.
[34] L. Melis, C. Song, E. De Cristofaro, and V. Shmatikov, "Exploiting unintended feature leakage in collaborative learning," in *Proc. IEEE Symp. Secur. Privacy*, 2019, pp. 691–706.

Authorized licensed use limited to: Universidad de Malaga. Downloaded on September 13,2026 at 16:03:57 UTC from IEEE Xplore. Restrictions apply.

[35] N. Liu, A. Nikitas, and S. Parkinson, "Exploring expert perceptions about the cyber security and privacy of connected and autonomous vehicles: A thematic analysis approach," *Transp. Res. Part F: Traffic Psychol. Behav.*, vol. 75, pp. 66–86, 2020.
[36] R. L. Rivest et al., "On data banks and privacy homomorphisms," *Found. Secure Comput.*, vol. 4, no. 11, pp. 169–180, 1978.
[37] C. Dwork, "Differential privacy," in *International Colloquium on Automata, Languages, and Programming*. Berlin, Germany: Springer, 2006, pp. 1–12.
[38] O. Goldreich, "Secure multi-party computation," *Manuscript. Preliminary Version*, vol. 78, no. 110, pp. 1–108, 1998.
[39] R. Sedar, C. Kalalas, F. Vázquez-Gallego, L. Alonso, and J. Alonso-Zarate, "A comprehensive survey of V2X cybersecurity mechanisms and future research paths," *IEEE Open J. Commun. Soc.*, vol. 4, pp. 325–391, 2023.
[40] B. Schoettle and M. Sivak, "A survey of public opinion about connected vehicles in the U.S., the U.K., and Australia," in *Proc. Int. Conf. Connected Veh. Expo*, 2014, pp. 687–692.
[41] C. Schmittner, Z. Ma, and T. Gruber, "Standardization challenges for safety and security of connected, automated and intelligent vehicles," in *Proc. Int. Conf. Connected Veh. Expo*, 2014, pp. 941–942.
[42] H. Guissouma, "Positionspapier: Chancen durch standardisierung und open-source-software im mobilitätssektor," 2024, doi: 10.5445/IR/1000168320.
[43] P. R. Donnellan, "The future of mobility—electric, autonomous, and shared vehicles," *IEEE Eng. Manag. Rev.*, vol. 46, no. 4, pp. 16–18, Fourthquarter 2018.
[44] M. Hossfeld, C. Ackermann, and C. Griffy-Brown, "A cyberphysical vehicle platform for the mobility of the future—creating new value networks and business models," *IEEE Eng. Manag. Rev.*, vol. 49, no. 4, pp. 99–107, Fourthquarter 2021.
[45] M. Hoßfeld, C. Ackermann, and T. Dietz, *Eine Redefinition des Plattformbegriffs*. Wiesbaden: Springer Fachmedien Wiesbaden, 2020, pp. 423–432.
[46] S. Nemet and D. Kukolj, "Analysing patents of start-ups in AI-based automotive industry," *Int. J. Technoentrepreneurship*, vol. 4, no. 1, p. 47, 2020. [Online]. Available: http://www.inderscience.com/link.php?id=108093
[47] F. Reichelt, D. Holder, and T. Maier, "The vehicle development process where engineering meets industrial design," *IEEE Eng. Manag. Rev.*, vol. 51, no. 4, pp. 102–123, Fourthquarter 2023.
[48] A. Haghighatkhah, A. Banijamali, O.-P. Pakanen, M. Oivo, and P. Kuvaja, "Automotive software engineering: A systematic mapping study," *J. Syst. Softw.*, vol. 128, pp. 25–55, 2017.
[49] R. Patil and V. Gudivada, "A review of current trends, techniques, and challenges in large language models (LLMS)," *Appl. Sci.*, vol. 14, no. 5, 2024, Art. no. 2074. [Online]. Available: https://www.mdpi.com/2076-3417/14/5/2074

Johannes Stümpfle received the bachelor's and master's degrees in electrical engineering from the University of Stuttgart, Stuttgart, Germany, in 2018 and 2022, respectively. He is currently working toward the Ph.D. degree. He is a Research Assistant with the Institute of Industrial Automation and Software Engineering, the University of Stuttgart. His research focuses on generative AI for software engineering, particularly its use in analyzing and restructuring large software systems to manage variability.

Authorized licensed use limited to: Universidad de Malaga. Downloaded on September 13,2026 at 16:03:57 UTC from IEEE Xplore. Restrictions apply.

Johannes Sigel received the bachelor's and master's degrees in electrical engineering from the University of Stuttgart, Stuttgart, Germany, in 2021 and 2023, respectively. He is currently working toward Ph.D. degree. He is a Research Assistant with the Institute of Industrial Automation and Software Engineering, University of Stuttgart. His areas of interest include the adaptation of multimodal automation systems and the interconnectivity of intelligent vehicles.

Matthias Weiß (Graduate Student Member, IEEE) received the bachelor's and master's degrees in electrical engineering from the University of Stuttgart, Stuttgart, Germany, in 2018 and 2021, respectively. He is currently working toward the Ph.D. degree and a Research Assistant with the Institute of Industrial Automation and Software Engineering, the University of Stuttgart. His research interests include the continuous analysis and optimization of distributed cloud/edge systems with the goal of reducing the operational complexity caused by frequent update deployments.

Baran Can Gül received the B.Sc. degree (Hons.) in electrical-electronics engineering from Yaşar University, Izmir, Türkiye, in 2019 and the M.Sc. degree in electrical engineering from the University of Stuttgart, Stuttgart, Germany, in 2022. He is currently working toward the Ph.D. degree with the Institute of Industrial Automation and Software Engineering, the University of Stuttgart, Germany. He is currently a Research Assistant. His current research focuses on federated learning and privacy-preserving machine learning, aimed at enhancing user safety and comfort through collaborative sensing.

Falk Dettinger received the bachelor's degree in electrical engineering from the DHBW Stuttgart, Stuttgart, Germany, in 2017. He received the master's degree from University of Stuttgart, Germany, in 2021. He is currently working toward the Ph.D. degree. He is a Research Assistant with the Institute of Industrial Automation and Software Engineering (IAS), University of Stuttgart. In addition, he is a Lecturer with the Faculty of Electrical Engineering, Baden-Wuerttemberg Cooperative State University (DHBW) Stuttgart. He joined the Broetje Automation GmbH as a Commissioning Engineer in the electrical engineering department. His areas of interest include function offloading between connected vehicles and distributed cloud/edge systems and intelligent function orchestration.

Nasser Jazdi (Senior Member, IEEE) received the Diploma degree in electrical engineering, the Ph.D. degree in remote diagnosis and maintenance of embedded systems from the University of Stuttgart, Stuttgart, Germany, in 1997 and 2003, respectively. In 2003, he joined the Institute of Industrial Automation and Software Engineering (IAS), University of Stuttgart as the Deputy Director, Researcher and Lecturer. He teaches four lectures: "Software Engineering (6 ECTS-CP)," "Reliability and Security of Industrial Automation Systems (3 ECTS-CP)," and "Dependability of Intelligent and Distributed Industrial Automation Systems (6 ECTS-CP)." Since 2019, he has been lecturing AI for Industrial Automation with the University of Anhui, China (6 ECTS). Up to now, he has led more than 37 industrial projects with companies such as Daimler AG, Siemens AG, Bosch GmbH, Porsche AG, ads-tec GmbH, Vector Informatik GmbH, Festo and BASF SE. His research interests include dynamic calculation of reliability using machine learning, software reliability in the context of IoT and artificial intelligence in industrial automation. He is a member of the VDE (The Association of German Engineers) and a member of VDI-GPP (Company Product and process design) Software Reliability Group. In 2009, he worked as an invited researcher with Prof. Zadeh at the University of California-Berkeley for three months. He has supervised more than 330 master, bachelor and student research projects. He is the author of 15 journals and 75 conference publications. He also has three book chapters. Dr. Jazdi is the Deputy Director of the Institute of Industrial Automation and Software Engineering at the University of Stuttgart and visiting Professor at the Anhui University in Hefei.

Authorized licensed use limited to: Universidad de Malaga. Downloaded on September 13,2026 at 16:03:57 UTC from IEEE Xplore. Restrictions apply.

Max Hoßfeld received the undergraduate and doctoral degrees from the University of Stuttgart, Stuttgart, Germany, in 2011 and 2016, respectively. He currently holds an associate professorship at the University of Quebec. He is the Managing Director of the InnovationCampus Mobility of the Future (ICM), a joint initiative of the University of Stuttgart and the Karlsruhe Institute of Technology. His research and teaching focus on advanced manufacturing, production management, and industrial value-creating networks. Before returning to academia, he held leadership roles in industry and applied research, including positions as Head of Engineering, Production Manager, and CTO. He is also a co-founder of three successful companies in the fields of robotics, artificial intelligence, and advanced manufacturing technologies.

Michael Weyrich studied at the Universities of Saarbrücken, Bochum, and London and received the Ph.D. degree in Multimediale Werkzeuge zur 3D-Planung im Maschinen- und Anlagenbau from RWTH Aachen, Aachen, Germany, in 1999. He holds an honorary doctorate from the Ukrainian University TUDon. He is the Head of the Institute for Automation and Software Engineering, University of Stuttgart. He researches and teaches in the area of industrial information technology for automation. His research interests include intelligent automation components, autonomous systems, and automation safety. Before joining the university, he worked in the multinational industry for 10 years, including four years in Asia. He is Chairman of the VDI/VDE Society for Measurement and Automation Technology and a member of the VDE Executive Board. He holds several positions, e.g., the Head of the VDI/VDE Technical Committee on testing networked systems, the Director of the Innovation Campus Mobility, industry chair of the International Federation of Control for “Computers for Control,” member of the program committee in the IEEE ETFA conferences as well as member of the editorial board of the journal ATP edition.

Authorized licensed use limited to: Universidad de Malaga. Downloaded on September 13,2026 at 16:03:57 UTC from IEEE Xplore. Restrictions apply.